LESSON PLAN

EHM (28)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	7	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Torque tightening procedure - calculation of torque, torque tightening practice and calibration procedure of torque wrench.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Torque tightening procedure - calculation of torque, torque tightening practice and calibration procedure of torque wrench. Practical on torque tightening procedure
5. Specific Objectives	28.1.3.2 Trainee will be able to explain about Torque Wrench and its usage.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Practical on torque tightening procedure; Safety precautions: Wear PPE, ensure zero tension before use; Instructor supervision: Monitor application technique and calibration verification steps.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a torque wrench?
(b) Why is accurate torque application critical in aircraft assembly?
(c) What are the common types of torque wrenches used in maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Torque wrench types and selection • Calculation of required torque values • Standard torque tightening procedure	L & Ds	PPT, IB	15 min
(b)	• Hands-on torque tightening practice • Calibration procedure of torque wrench • Verification of applied torque values	L & Ds	PPT, PDF	20 min
(c)	• Common errors in torque application • Maintenance and storage of torque tools • Documentation of calibration records	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the step-by-step procedure for calibrating a torque wrench.
(b) Describe the consequences of under-torquing or over-torquing a fastener.
(c) How do you verify that a torque wrench is functioning correctly before use?

12. Home Assignment and Follow Up	Prepare a brief report on the importance of torque wrench calibration for the next lesson on Riveting and De-riveting (LP 02).
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Riveting and De-riveting: Types of rivets, rivet parameters calculation, standard riveting procedure, de-riveting of loose rivets, heat treatment of rivets, inspection procedure.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min